

Group 9 Assignment Report

An Evolutionary Algorithm for the Travelling Salesman Problem

Christos Zampounis Mehmet Cihan Yalcin Péter Dávid Hajdu
Pierre Kasparian Prajval Parthiban

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1 Introduction

The goal of this assignment is to design, implement and experimentally evaluate an evolutionary algorithm for a combinatorial optimisation problem. We chose the Travelling Salesman Problem, whose discrete, permutation-based search space and relevance to routing and logistics make it a natural and challenging testbed for evolutionary computation. This report presents our problem formulation, the evolutionary approach and operators we developed, and an experimental analysis of its solution quality and convergence behaviour.

2 The Travelling Salesman Problem

2.1 Mathematical Formulation

The Travelling Salesman Problem (TSP) is a classical combinatorial optimisation problem that, despite its simple formulation, is difficult to solve: given a set of N cities and the distances between them, the task is to find the shortest closed tour that visits every city exactly once. The problem is NP-hard and no exact algorithm can solve it in polynomial time. Following the formulation of Potvin [1], we model the problem on a complete weighted graph $G = (V, E)$, where $V = \{1, \dots, N\}$ is the set of cities and d_{ij} is the distance between cities i and j . The decision variables are $x_{ij} \in \{0, 1\}$, where $x_{ij} = 1$ if and only if the tour travels directly from city i to city j . The objective is to minimise the total travelled distance,

$$\text{Minimise} \quad \sum_{i=1}^N \sum_{j=1}^N d_{ij} x_{ij}, \quad (1)$$

subject to the degree constraints

$$\sum_{j=1}^N x_{ij} = 1, \quad i = 1, \dots, N, \quad \sum_{i=1}^N x_{ij} = 1, \quad j = 1, \dots, N, \quad (2)$$

which require every city to have exactly one incoming and one outgoing edge, and the subtour-elimination constraints

$$\sum_{i,j \in S} x_{ij} \leq |S| - 1, \quad \forall S \subset V, \quad 2 \leq |S| \leq N - 1, \quad (3)$$

which forbid partial cycles over any proper subset S of the cities. Without constraints (3), the model reduces to the assignment problem and its optimal solution generically consists of several disconnected subtours rather than a single tour. For symmetric instances, such as

the Euclidean TSP (ETSP) considered in this report, the optimal solution of the assignment-problem relaxation typically contains many two-vertex subtours; such instances are therefore better tackled by specialised algorithms that exploit this structure [1].

Because an evolutionary algorithm operates on orderings rather than on binary edge variables, it is convenient to state the equivalent tour-as-permutation formulation. A candidate solution is a permutation $\pi = (\pi_1, \dots, \pi_N)$ of the cities, and its cost is the length of the cycle that visits them in that order,

$$C(\pi) = \sum_{k=1}^{N-1} d_{\pi_k, \pi_{k+1}} + d_{\pi_N, \pi_1}. \quad (4)$$

The evolutionary algorithm therefore seeks a permutation minimising Equation (4); this representation is the one adopted throughout this report.

3 Algorithm

4 Experimental part

5 Results and Analysis

6 Conclusions

References

- [1] J.-Y. Potvin, “Genetic algorithms for the traveling salesman problem,” *Annals of Operations Research*, vol. 63, no. 3, pp. 339–370, 1996. <https://doi.org/10.1007/BF02125403>.